

Slide 1: Title

Team Name: Syntheon

Team Leader Name: Saurav Soni

Problem Statement: Data & AI Challenge — Intelligent Candidate Discovery

Solution: RedrobRank Agentic Recruiter OS

Slide 2: Solution Overview

RedrobRank is a deterministic CPU-only TalentOps agent that ranks 100K candidates into a validator-compliant Top-100 shortlist.

Differentiators:

- JD-aware evidence extraction
- production/evaluation/retrieval-first scoring
- Redrob behavioral signal modulation
- honeypot/risk filtering
- no LLM/API calls
- fully reproducible

Slide 3: JD Understanding & Candidate Evaluation

JD requirements:

- 5–9 YOE senior AI engineer
- retrieval/ranking/LLM/fine-tuning
- production shipping
- evaluation infrastructure
- recruiter/product workflow mindset
- Pune/Noida/tier-1 India/relocation

Signals: title + career history, actual skills, production terms, evaluation terms,
Redrob signals, risk flags

Beyond keyword matching: penalizes non-technical roles, checks production depth,
checks evaluation language, acknowledges concerns

Slide 4: Ranking Methodology

Pipeline:

1. Stream JSONL
2. Extract text from profile/career/skills/signals
3. Score by weighted evidence
4. Apply risk penalties
5. Sort score desc + candidate_id tie-break
6. Generate grounded reasoning

Weights:

AI/retrieval/ranking, production, evaluation, systems, YOE,
location, behavioral signals, risk.

Slide 5: Explainability & Data Validation

Each reasoning references:

- actual title
- YOE
- location
- matched skills
- production/eval count
- Redrob response/open/relocation signal

Hallucination prevention: no LLM generation, only deterministic extracted evidence,
validator + grounding audit

Suspicious profiles: honeypot penalties, role mismatch penalties,
pure research / manager-only checks

Slide 6: End-to-End Workflow

JD -> Signal Dictionary

- > Candidate JSONL Stream

- > Feature Extractor

- > Risk Detector

- > Weighted Ranker

- > Top-100 CSV

- > Validator

- > Dashboard/Sandbox

Slide 7: System Architecture

CLI Ranker:

- JD config, Candidate stream reader, Feature extractor, Scoring engine
- Risk engine, Reasoning generator, CSV writer, report generator

React Dashboard: reads reports/*.json

Streamlit Sandbox: sample upload, sample rank, CSV download

No network / no LLM / no GPU

Slide 8: Results & Performance

- 100,000 candidates processed
- runtime under 5 minutes CPU-only
- final CSV validates
- 100 ranked rows
- no raw dataset committed
- no external APIs
- top-100 reasoning generated
- trap/honeypot logic included

Slide 9: Technologies Used

- Python
- deterministic JSONL streaming
- custom rule-based scoring
- pytest / coverage / ruff
- Bandit / Semgrep / pip-audit / npm audit
- React + Vite dashboard
- Streamlit sandbox
- Sentrux structural quality scan
- GitHub public repo

Slide 10: Submission Assets

GitHub: <https://github.com/Sauravssoni/Screeener>

CSV: submissions/submission.csv

Deck: docs/RedrobRank_Official_Submission_Deck.pdf

Methodology: docs/RedrobRank_Methodology.pdf

Run: `python3 rank.py --candidates data/candidates.jsonl --out submissions/submission.csv`

Validate: `python3 validate_submission.py submissions/submission.csv`

Sandbox: `streamlit run sandbox_app.py`

Slide 11: Thank You

Thank You